Experiment 10 **Digital Electronics** **Monsoon 2018**

Simple Sequential Circuits using Verilog

This experiment consists of writing simple Verilog programs to simulate and verify the logic of various kinds of sequential circuits.

1. Write a Verilog code for generating a clock of time period 10 ns. The module to be written is expected to have just one output ***clk*** and no other inputs.

2. Write the Verilog code for a positive edge D Flip-Flop having an input ***D***, an output ***Q*** and a synchronous ***reset*** input. Verify the functionality of the D Flip-Flop by writing a test-bench. The test-bench should drive the ***D*** and ***reset*** input to some arbitrary varying signal and ***clk*** should be driven by the instantiated clock generator block created in **step 1**. Observe the output ***Q*** and its dependency on ***clk***, ***reset*** and ***D***.

3. Convert the D flip-flop and test-bench created in **step 2** into an asynchronous-reset D flip-flop. Show the effect of asynchronous reset on ***Q*** in the absence of clock edge.

4. Write a Verilog code for a negative edge-triggered D Flip-Flop with asynchronous reset. Instantiate this design in the same test-bench created in **step 3** to verify its behaviour.

5. Write a Verilog code for a 4-bit counter which counts from 0 to 15. The count should increment on every clock and should go back to 0000 after 15 (1111). Simulate it using a clock given by the the ***clk*** output from the test-bench created in **step 2**.